Hasselt University

Project Learning from Data

Determinants of Residential Energy Consumption in Belgium

Group 1

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Group-specific rows removed from raw dataset: 1, 11, 40

1 Introduction

This report assesses the determinants of annual residential electricity consumption in Belgium using a cross-sectional sample of 2,500 households. Four research questions guide the analysis: (Q1) identifying the best predictive model; (Q2) quantifying the consumption difference between EPC label A and labels B–F; (Q3) assessing the direct effect of household income on consumption; and (Q4) testing whether the effect of the EPC label varies with household occupancy. Variables `roof_color` and `pet_ownership` were excluded a priori as they represent neither functional nor socio-economic drivers of energy demand (Vasseur & Kemp, 2015). This is confirmed by bivariate screenings (Appendix Figure 5), where consumption remain the same across all categories.

2 Data and Preprocessing

After removing the rows following the instructions of the assignment (group-specific rows removed from the raw dataset: 1, 11, 40) and *a priori* excluded variables, `epc_label` (levels A–F; reference: A) and `building_era` (Pre-1970, 1970–2000, Post-2000) were converted to factors. Fifteen incomplete rows ($<$ 1%) were removed via listwise deletion, yielding n = 2,482.

Outlier analysis. A 3.0 × IQR upper fence identified seven extreme values of `annual_kwh` (Tukey, 1977). Five showed consumption in the millions of kWh, which is physically impossible for residential households of their reported size, occupancy, and EPC class, as shown in Table 1, and were removed as considered data entry errors. Two high values (16,395 and 17,197 kWh) were retained as internally consistent: large homes (268–304 m^2), seven occupants, Pre-1970, EPC E/F. The final analytical dataset contains **2,477 observations**.

Table 1: Extreme observations identified by 3.0 × IQR rule.

ID	annual_kwh	m^2	Occupancy	EPC	Decision
Obs. A	2,900,000	112	2	A	Removed
Obs. B	3,400,000	100	1	B	Removed
Obs. C	4,100,000	156	4	C	Removed
Obs. D	5,400,000	130	3	D	Removed
Obs. E	2,100,000	142	2	E	Removed
Obs. F	16,395	304	7	E	Retained
Obs. G	17,197	268	7	F	Retained

Variable exclusions. A chi-square test between `epc_label` and `building_era` yielded $p \approx 0$, with zero category overlap in the contingency table (Pre-1970 maps exclusively to C–F; Post-2000 to A–B): perfect collinearity (Appendix Table 5). `epc_label` was retained as it appears directly in Q2 and Q4. `dist_to_brussels` was excluded due to zero Pearson correlation with `annual_kwh` (r = 0.00) shown in Table 2.

2.1 Descriptive Statistics

Table 7 in Appendix summarizes the descriptive statistics for $n = 2,477$. The average annual energy consumption is 5,873.90 kWh (SD $= 2,224.00$), while living space averages 158.30 m^2. Occupancy centers at 3.99 residents, and the mean household income is €48,861.00, ranging from €8,488.0 to €206,368.0.

2.2 Key Correlations

There is strong correlation between income_euro and sq_meters (r = 0.72) explains why income is absorbed by the living area in the model. Zero correlation between dist_to_brussels and annual_kwh (r = 0.00) justifies its exclusion.

Table 2: Pearson correlation matrix

	kwh	sq_m	occup.	income	dist
annual_kwh	1.00	0.38	0.76	0.29	0.00
sq_meters	0.38	1.00	-0.02	0.72	0.02
occupancy	0.76	-0.02	1.00	0.00	0.02
income_euro	0.29	0.72	0.00	1.00	-0.01
dist_brussels	0.00	0.02	0.02	-0.01	1.00

3 Statistical Methods

Forward variable selection was applied to the modeling dataset in the order specified in the Statistical Analysis Plan (SAP): `sq_meters` → `epc_label` → `occupancy_count` → `income_euro` → `epc_label:occupancy_count`. A term was retained only if it was statistically significant ($p < 0.05$) **and** reduced the model AIC. After each step, the following diagnostics were performed as described in the SAP: residuals vs. fitted plots (linearity and homoscedasticity), Q-Q plots plus Kolmogorov-Smirnov test (normality), and GVIF (multicollinearity). Where heteroscedasticity was detected, a log transformation of the outcome was applied and the selection repeated. The modeling strategy and diagnostic procedures follow standard linear regression practices (James et al., 2021).

Because the SAP explicitly planned three distinct hypothesis tests on the final model (Q2, Q3, and Q4), Holm-Bonferroni correction was applied to control the familywise error rate (FWER) at $\alpha = 0.05$. The Holm-Bonferroni stepwise procedure was chosen because it strongly controls FWER under any dependence structure, is more powerful than the classical Bonferroni correction, and remains valid for the small number of tests performed in our analysis (Holm, 1979).

4 Results

4.1 Model Selection and Diagnostics

Forward selection on the original scale produced a model with pronounced heteroscedasticity and non-normal residuals (Figures 1 and 2, left panels). A log transformation of `annual_kwh` resolved both violations (right panels), justifying the transformation on the diagnostic grounds rather than EDA alone.

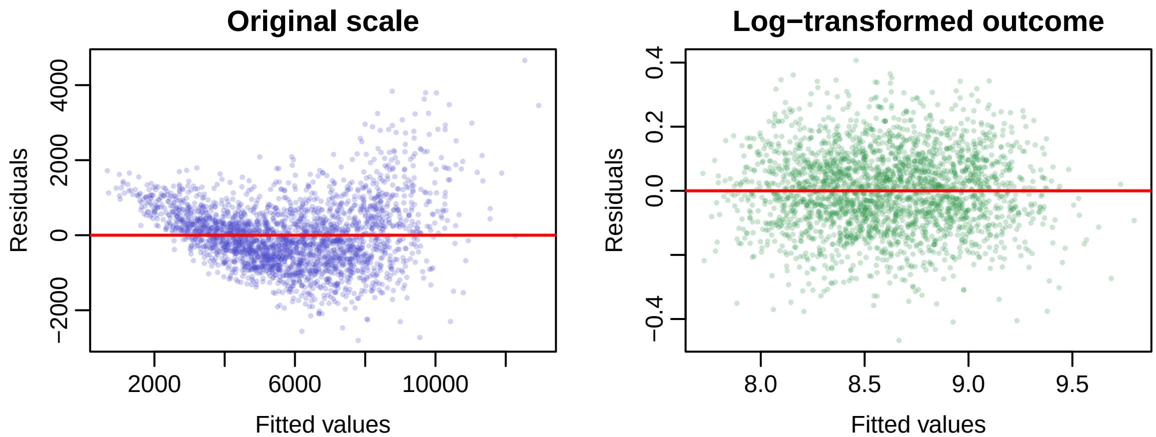

Figure 1: Residuals vs. Fitted before (left) and after (right) log transformation. The fan-shaped pattern on the left confirms heteroscedasticity; the log scale resolves it.

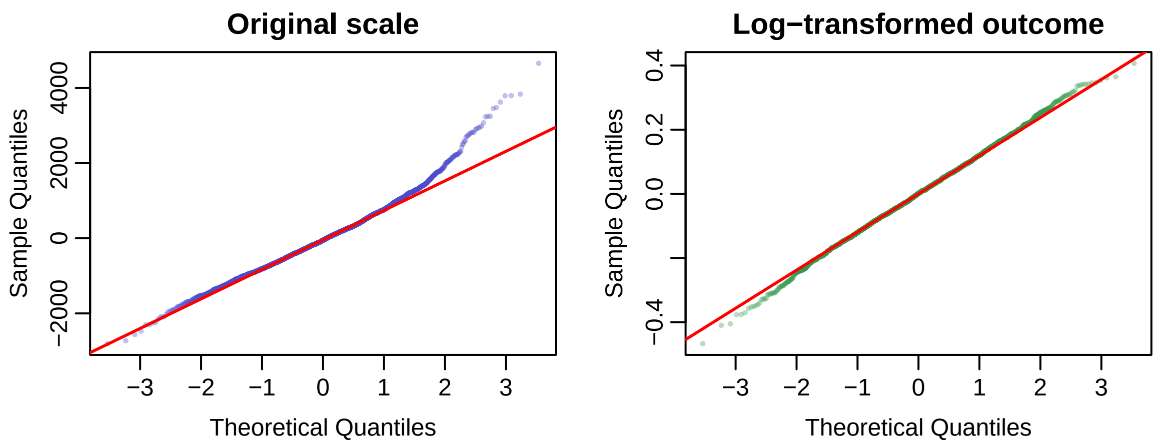

Figure 2: Q-Q plots of residuals before (left) and after (right) log transformation. Normality is substantially improved after transformation (Kolmogorov-Smirnov test on final model: p = 0.94).

Table 3: Forward selection on log(annual_kwh). $\star$ = selected final model.

Model	AIC	R^2	Adj. R^2	RMSE	Decision
M1: sq_meters	1749.4	0.142	0.142	0.3440	enters
M2: + epc_label	1435.2	0.247	0.246	0.3222	enters
M3: + occupancy_count $\star$	-3349.3	0.891	0.891	0.1226	**enters**
M4: + income_euro	-3347.7	0.891	0.891	0.1226	excluded ($p = 0.534$)
M5: + epc_label:occupancy	-3344.7	0.891	0.891	0.1225	excluded (all $p > 0.05$)

The final model is:

$$\widehat{\log(\text{annual\_kwh})} = \beta_0 + \beta_1 \cdot \text{sq\_meters} + \sum_{k \in \{B,\ldots,F\}} \beta_k \cdot \mathbf{1}[\text{epc} = k] + \beta_7 \cdot \text{occupancy\_count}$$

with $R^2 = 0.891$, Adj. $R^2 = 0.891$, $AIC = -3349.3$, $RMSE = 0.123$ (log scale). All GVIF values were below 2.4, indicating no problematic multicollinearity.

4.2 Model Parameters

Table 4: Final model coefficients on log scale. Reference category: EPC A.

	Term	Estimate	SE	95% CI Low	95% CI High	p-value
(Intercept)	Intercept	7.2106	0.0130	7.1851	7.2362	< 0.001
sq_meters	sq_meters	0.0040	0.0001	0.0038	0.0041	< 0.001
epc_labelB	epc B	0.0869	0.0079	0.0714	0.1023	< 0.001
epc_labelC	epc C	0.1583	0.0075	0.1436	0.1730	< 0.001
epc_labelD	epc D	0.2414	0.0081	0.2254	0.2574	< 0.001
epc_labelE	epc E	0.3249	0.0085	0.3082	0.3415	< 0.001
epc_labelF	epc F	0.4025	0.0095	0.3839	0.4212	< 0.001
occupancy_count	occupancy_count	0.1511	0.0013	0.1486	0.1535	< 0.001

All parameters are highly significant ($p < 0.001$). On the original kWh scale (via back-transformation): each additional m^2 increases consumption by 0.4%; each additional resident increases consumption by **16.3%**, the dominant predictor. Relative to EPC A, the consumption gradient is monotonic: EPC B +9.1%, EPC C +17.1%, EPC D +27.3%, EPC E +38.4%, EPC F +**50.0%**. The intercept (exp(7.211) $\approx$ 1,353 kWh) represents a theoretical EPC A home with zero area and occupancy, and is not directly interpretable.

4.3 Research Questions

Q1: Best predictive model. The model `log(annual_kwh) ~ sq_meters + epc_label + occupancy_count` is selected on grounds of parsimony: lowest AIC ($-3{,}349$) and highest adjusted R^2 (0.891) among all candidates. Adding `income_euro` or the interaction term improves neither metric.

Q2: EPC A vs. average B–F. Figure 3 shows a clear monotonic consumption gradient across EPC levels. Formally, EPC A homes consume on average **21.6% less** than the mean of B–F homes (ratio = 0.784, 95% CI: 0.775–0.794; Holm-adjusted $p < 0.001$), controlling for living area and occupancy.

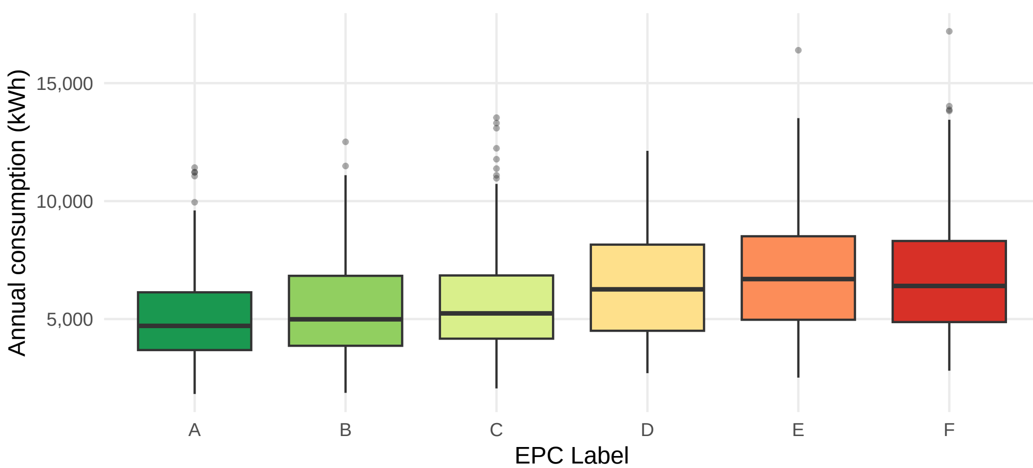

Figure 3: Annual energy consumption (kWh) by EPC label. The monotonic increase from A to F supports the EPC label effect and contextualises the Q2 contrast.

Q3: Income effect. The direct effect of income is negligible and non-significant: per €10,000 increase, ratio = 1.001 (95% CI: 0.998–1.004; Holm-adjusted $p = 0.534$). The apparent wealth effect operates indirectly through living area ($r = 0.72$): once area is controlled, income adds no explanatory power.

Q4: EPC × occupancy interaction. The interaction did not meet the entry criterion (all five terms $p > 0.05$, as shown in Table 8 in the Appendix). Figure 4 confirms that lines are near-perfectly parallel across all EPC levels, indicating that the EPC effect is consistent regardless of household size.

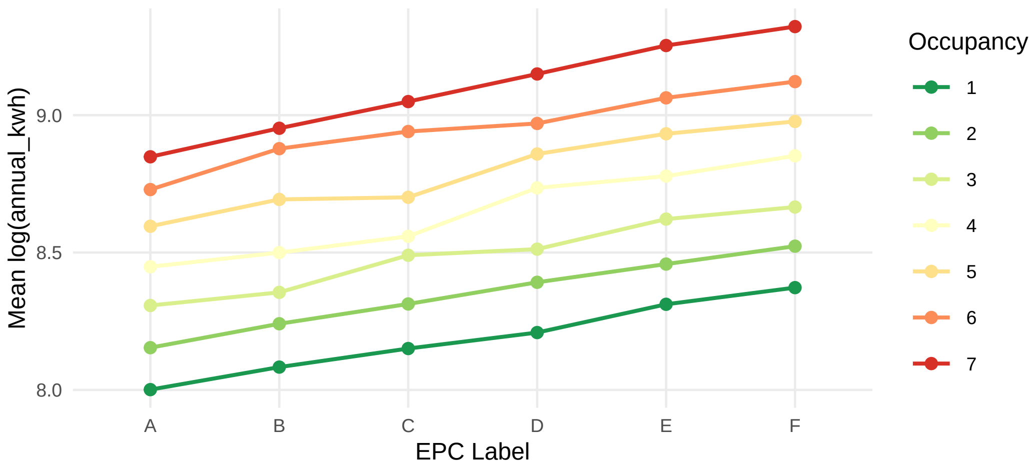

Figure 4: Interaction plot: mean log(annual_kwh) by EPC label for each occupancy level. Near-perfectly parallel lines confirm the absence of a significant interaction.

5 Ethical Considerations and Stakeholder Impact

This project worked with data containing sensitive information such as income, occupancy, and location of households. Although the data is anonymized, there is a risk of re-identification through a combination of these variables; therefore, this data should be handled with care and responsibility, adhering to the principles of data governance and in compliance with the General Data Protection Regulation law framework. Additionally, this analysis identified that the EPC label is an important predictor of energy consumption; therefore, one ethical way to use these results is to help people through government renovation subsidies and discounted energy rates, focusing on fixing the buildings themselves. Finally, to maintain professional integrity, this analysis explicitly advises against using this data in private-sector algorithms, for example, for institutions like banks or insurance companies, using home energy efficiency ratings to judge customers would introduce an unfair bias against lower-income renters who lack the means to improve their homes. The use of this information for commercial exploitation must therefore be prevented. If predatory real estate developers or contractors were to use these records to target inefficient homes, they could pressure vulnerable residents or buy cheap housing for profit.

6 Conclusion

Residential energy consumption in Belgium is primarily driven by household occupancy, EPC label, and living area. The final model `log(annual_kwh) ~ sq_meters + epc_label + occupancy_count`, explains 89.1% of variance and satisfies all linear model assumptions after log transformation. EPC A homes consume 21.6% less than the B–F average. Income has no direct effect once the living area is controlled. The EPC–occupancy interaction is not statistically significant. These findings support targeted energy efficiency policies, but equitable implementation requires attention to housing access, data privacy, and the indirect pathways through which socioeconomic status shapes residential energy consumption.

References

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

Vasseur, V., & Kemp, R. (2015). The adoption of energy efficiency measures by Dutch households: On the role of relevant induced costs and innovativeness. Energy Policy, 77, 68-78

Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning. Springer.

AI Use Transparency Statement

During the preparation of this report, we used generative AI as an auxiliary writing-support tool. AI was used to review the draft text for grammatical accuracy and clarity of expression and to provide feedback on paragraph structure, transitions, and coherence. No text was copied verbatim from AI outputs into the final submission without substantial human review.

Appendix

A Variable Exclusion: Bivariate Screening

The main report excludes *roof_color* and *pet_ownership* a priori on theoretical grounds. The boxplots below provide empirical support: both variables show virtually identical distributions of `annual_kwh` across their categories, confirming no meaningful association with the outcome.

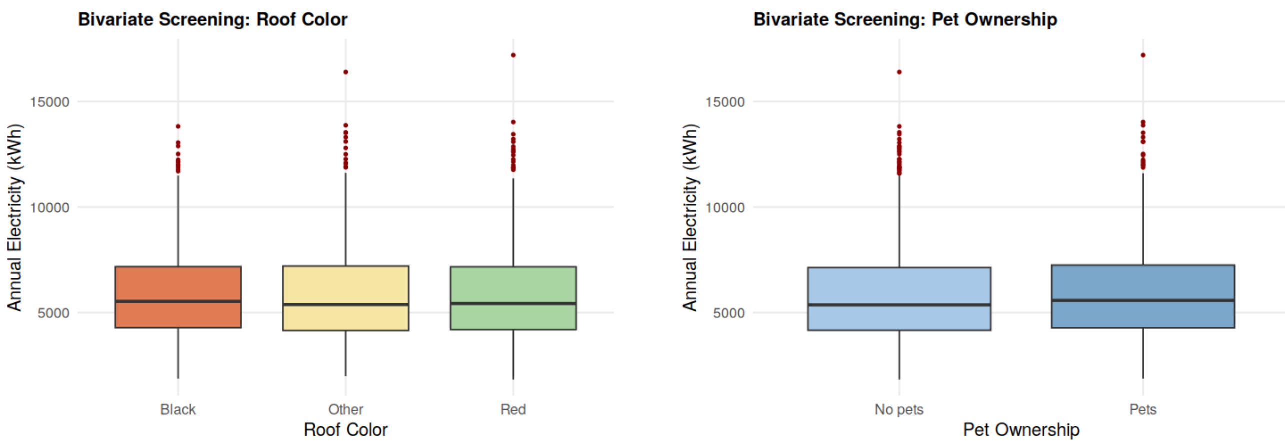

Figure 5: Annual electricity consumption by roof color (left) and pet ownership (right). Medians and interquartile ranges are nearly identical across all categories, supporting their a priori exclusion.

B `building_era` Exclusion: Contingency Table

The main report states that a chi-square test between `epc_label` and `building_era` revealed perfect collinearity ($p \approx 0$, zero category overlap). The table below makes this explicit: Post-2000 buildings map exclusively to EPC A–B; Pre-1970 and 1970–2000 buildings map exclusively to EPC C–F. `epc_label` was retained as it directly answers Q2 and Q4.

Table 5: Contingency table of `building_era` × `epc_label` ($n = 2{,}477$). Zero cells confirm perfect separation. Chi-square: $X^2 = 3311.6$, $df = 10$, $p < 2.2 \times 10^{-16}$.

	A	B	C	D	E	F
Pre-1970	0	0	96	149	336	239
1970–2000	0	180	431	237	0	0
Post-2000	553	256	0	0	0	0

C Descriptive Statistics

Table 6 summarises the key continuous variables in the final analytical dataset ($n = 2{,}477$) after removal of group-specific rows, listwise deletion of missing values, and exclusion of five confirmed data-entry errors.

Table 6: Descriptive statistics for continuous variables ($n = 2{,}477$).

Variable	Mean	SD	Min	Max
annual_kwh (kWh)	5873.90	2224.00	1824.6	17197.1
sq_meters (m^2)	158.30	36.50	60.3	341.3
occupancy_count	3.99	1.98	1.0	7.0
income_euro (€)	48861.00	22537.00	8488.0	206368.0

D Full GVIF Table

The main report states all GVIF values were below 2.4. The comparison column is $\text{GVIF}^{1/(2 \cdot Df)}$, with a cutoff of 2.24 (equivalent to VIF < 5).

Table 7: GVIF computed using `vif()` from the `car` package.

Variable	GVIF	Df	$\text{GVIF}^{1/(2 \cdot Df)}$	Verdict
`sq_meters`	1.002	1	1.001	< 2.24 ✓
`epc_label`	1.007	5	1.001	< 2.24 ✓
`occupancy_count`	1.006	1	1.003	< 2.24 ✓

E Residuals vs Leverage Plots

The Residuals vs Leverage plot checks whether any single observation has disproportionate influence on the model coefficients via Cook's distance. Points outside the dashed contours would require investigation.

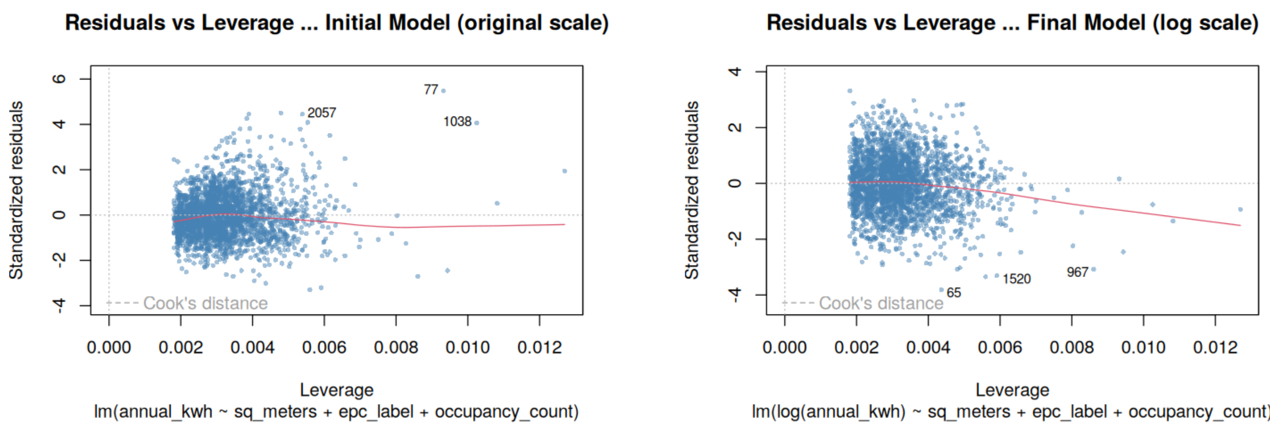

Figure 6: Residuals vs Leverage for original scale (left) and log scale (right).

F Model with EPC × occupancy interaction

Model M5 coefficients: `log(annual_kwh)` ∼ `sq_meters + epc_label + occupancy_count + epc_label:occupancy_count`.
The five interaction terms (shaded) all have $p > 0.05$, confirming the absence of a significant EPC × occupancy interaction.

Table 8: Model M5 estimated coefficient

Term	**Estimate**	**SE**	***t*-value**	***p*-value**
Intercept	7.2190	0.0165	437.326	< 0.001
`sq_meters`	0.0040	0.0001	58.366	< 0.001
EPC B	0.0746	0.0179	4.175	< 0.001
EPC C	0.1643	0.0169	9.714	< 0.001
EPC D	0.2124	0.0188	11.281	< 0.001
EPC E	0.3166	0.0196	16.160	< 0.001
EPC F	0.3866	0.0209	18.535	< 0.001
`occupancy_count`	0.1491	0.0027	56.092	< 0.001
EPC B × occupancy	0.0030	0.0040	0.758	0.449
EPC C × occupancy	−0.0017	0.0038	−0.442	0.659
EPC D × occupancy	0.0071	0.0041	1.709	0.088
EPC E × occupancy	0.0020	0.0043	0.464	0.643
EPC F × occupancy	0.0041	0.0049	0.844	0.399

Reference category: EPC A. $n = 2{,}477$. Adj. $R^2 = 0.891$.